

# CareCompanion

AI-powered visit prep and note digitization, built for neurodivergent patients

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# Medical visits are hard to prepare for, and harder to remember

1

## Prepping is overwhelming

Organizing symptoms, history, and questions under time pressure taxes executive function, especially for neurodivergent patients.

2

## Instructions are illegible

Handwritten after-visit notes and quick instructions are still common, and easy to misread once you're home.

3

## Medication tracking is high-stakes

Many neurodivergent patients manage controlled-substance medications (e.g. ADHD stimulants), where careful documentation matters more.



# E-prescribing digitized the drug order. Nobody digitized the conversation.

94%

of U.S. prescriptions are now e-prescribed

## What exists today

## What's missing

Patient portals show the official e-prescribed drug order

Nothing digitizes the handwritten notes, quick instructions, and side-conversation given during a visit

General AI chat tools can summarize text if you type it in

Nothing purpose-built for structuring a visit before it happens, or reading a doctor's actual handwriting after

Prescription-reader tools target pharmacists at the counter

No consumer-facing tool for the patient's own experience, before and after



# Built first for neurodivergent patients, useful for everyone

- Executive-function load makes unstructured prep (“just remember your symptoms”) especially hard
- Many manage controlled-substance medications (e.g. ADHD stimulants), where documentation and follow-through carry extra weight
- Benefit disproportionately from structure: guided prompts instead of a blank page, checklists instead of memory

## Why this framing matters

A tool built to reduce cognitive load for the population that needs it most also reduces friction for every patient. It's the same reason curb cuts help everyone, not just wheelchair users.



# Two-sided pipeline: before the visit, and after

## PRE-VISIT

- 1 Patient chats through symptoms and worries
- 2 Structuring engine organizes into chief complaint, timeline, and priority questions
- 3 Output: a one-page agenda to bring to the visit

## POST-VISIT

- 1 Photograph a handwritten note or after-visit instructions
- 2 Fine-tuned handwriting model reads it; structuring step sorts it into fields
- 3 Output: a clear checklist, flagged items marked for human review

# Three components, three different jobs

1

## Classifier : Open CV

Image pre-processing : Isolate the handwritten lines and visual features from the notes.

2

## Fine-tuned TrOCR (transformer)

An embedding + dense-layer network trained to sort recognized text into fields, such as medication, dosage, frequency, and follow-up, replacing simple keyword matching. TrOCR, a vision-transformer handwriting reader, fine-tuned on a real handwritten-prescription dataset rather than used out of the box.

3

## Conversational intake (pre-visit)

A structured chat flow that turns a patient's free-form worries into an organized pre-visit agenda. Conversations with LLM to organize patients thoughts



# Inside TrOCR: a vision encoder and a text decoder

## Encoder

A Vision Transformer, initialized from BEiT. The image is split into 16x16 pixel patches and turned into a sequence the transformer can read, the same way a sentence is a sequence of words.

## Decoder

A text Transformer, initialized from RoBERTa. It reads the encoder's output and generates the recognized text one piece at a time, using language patterns it already knows to make better guesses on messy handwriting.

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*Originally pretrained on hundreds of millions of printed and synthetic handwritten text lines, then fine-tuned on the IAM Handwriting Database to produce the general handwritten-text checkpoint we started from, before fine-tuning it further on prescription-specific handwriting ourselves.*

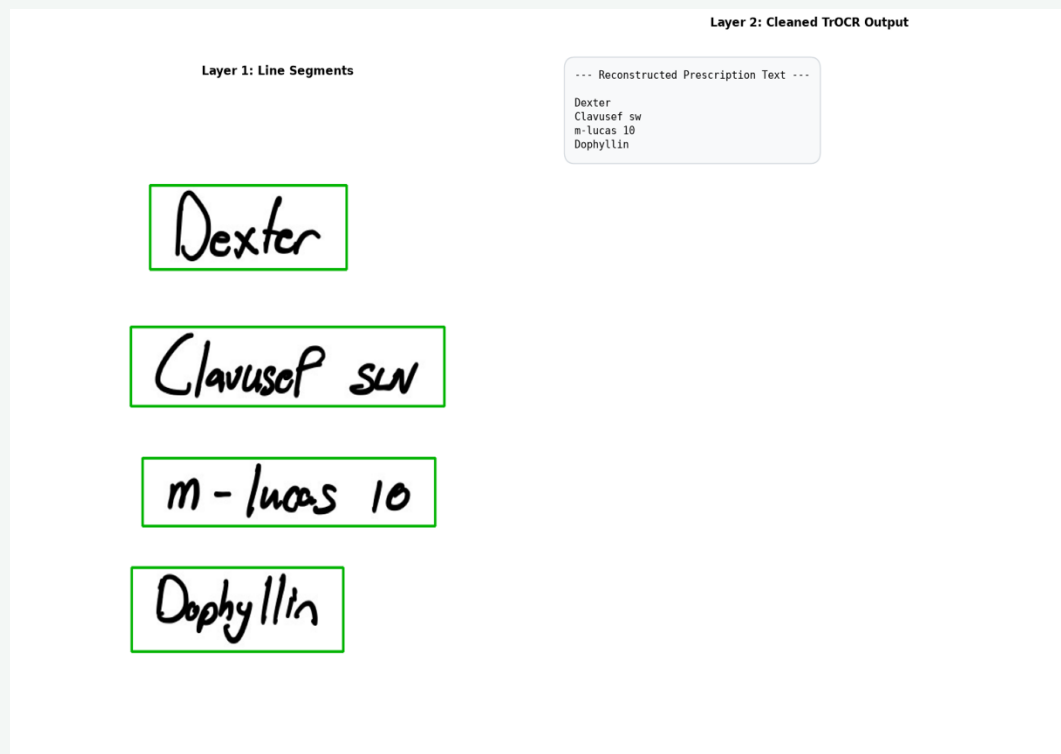


# Before the model ever sees it: the OpenCV pipeline

- 1 Convert to grayscale and normalize contrast
- 2 Apply thresholding to separate ink from background
- 3 Use morphological operations to merge letters into line-shaped blobs
- 4 Detect contours around each blob to draw a bounding box per line

*This is fixed, rule-based image processing, not a trained model. It's what hands the model clean, cropped lines to read, instead of one messy full-page photo.*

For fun: this step on a sample scrawl





# A locally-run LLM, kept strictly on task

The conversational intake component runs Llama 3.2 (3B) locally via Ollama, not a hosted API. Rather than a general-purpose chatbot, it's given a narrow role through its system prompt, one for before the visit and one for after, and a required output schema, so its output is structured, not free-form conversation.

## Pre-appointment role

*"You are an organizational clinical communication assistant. Your task is strictly organizational (never diagnose or recommend treatments). Extract the patient's concerns into the required schema fields."*

## Post-appointment role

*"You are an appointment follow-up assistant. Extract actionable instructions from clinical notes into the required schema fields."*



# Trained and measured, not assumed

## Handwriting model

Dataset: HuggingFace dataset , a public handwritten-prescription word-image dataset

### Baseline (zero-shot)

### High eval loss

Out-of-the-box model, not tuned to prescription handwriting

### Fine-tuned

### Lower eval loss

After fine-tuning on prescription-specific handwriting

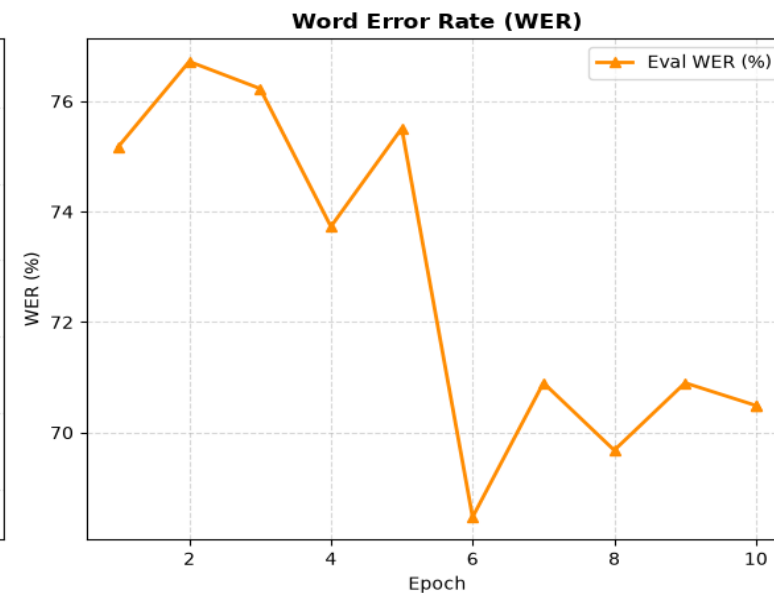
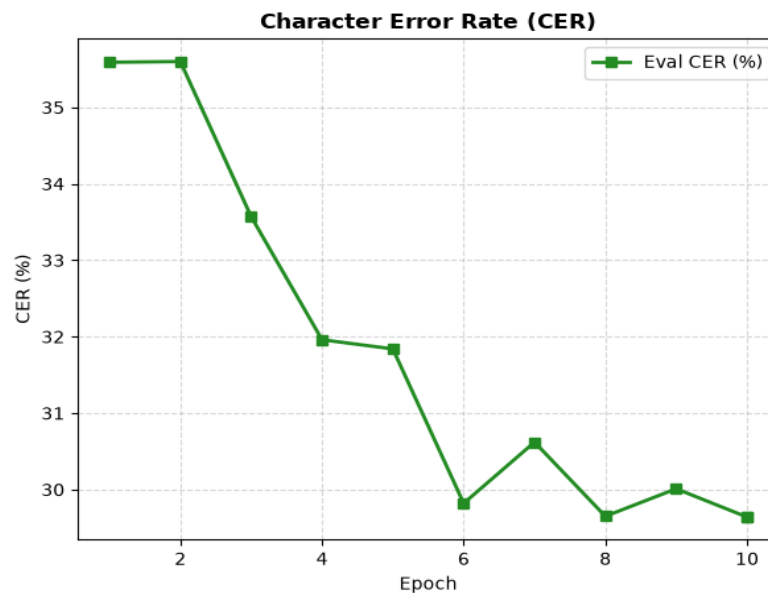
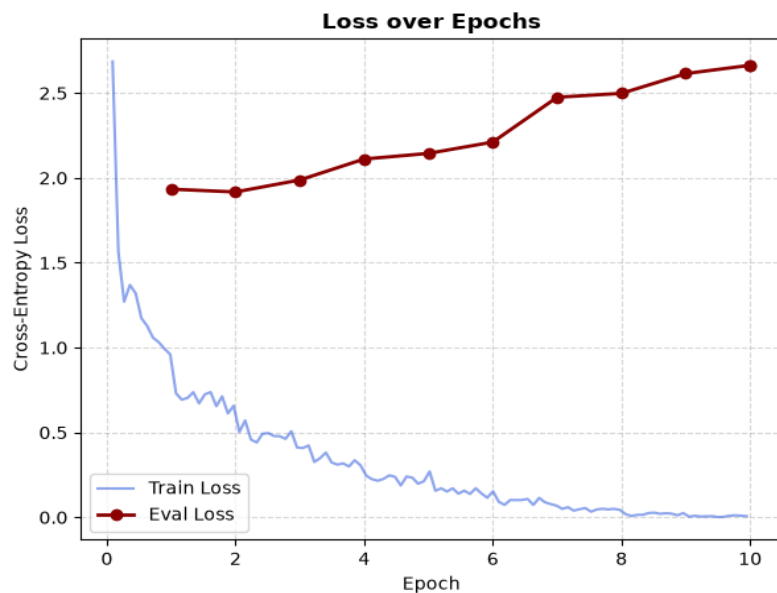
## Field classifier

Dataset: custom-labeled phrase set across categories (medication, dosage, frequency, and more)

90 / 10 train-test split, trained from scratch with an embedding layer, a hidden layer, and dropout for regularization. Loss and accuracy tracked across training epochs.



# Measured every epoch, not just at the end



29.6%

best Character Error Rate (epoch 8), down from 35.6% at epoch 1

~70%

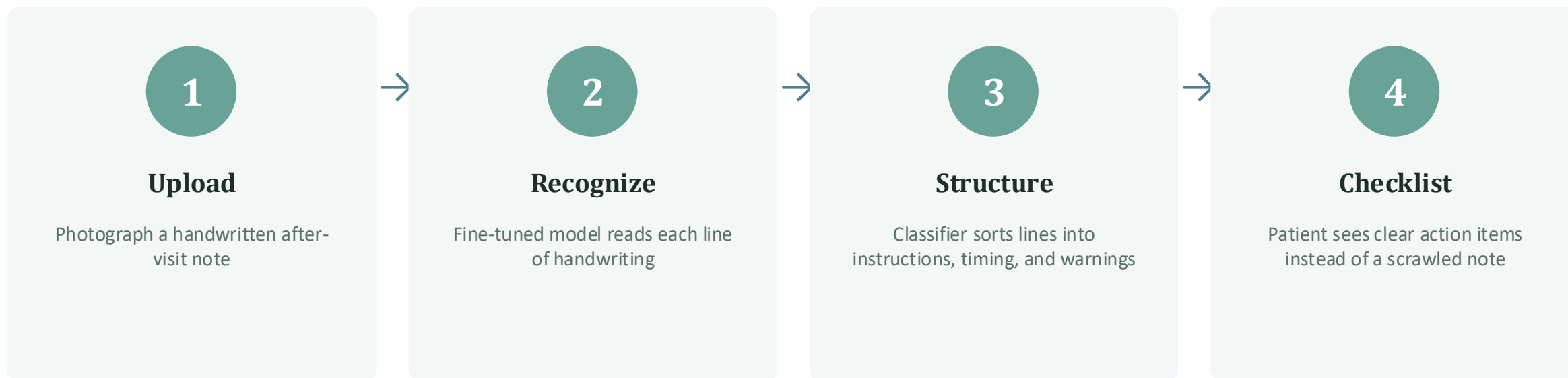
Word Error Rate stayed high and volatile across training

## Honest read

Validation loss rises even as training loss falls, a sign of overfitting on a small dataset. We report the best checkpoint, not just the last one.



# The confusing after-visit note



*Example: "ice 20min 3x/day, f/u 2wks, avoid NSAIDs" → "Ice affected area, 20 minutes, 3x daily · Follow-up in 2 weeks · Avoid NSAIDs"*



# The patient's own notes, taken in the moment

Unlike a clinician's note, a patient's own scrawled notes don't follow a predictable format, so the structuring step works differently here.

## Things to remember

"Dr. said not urgent"  
"Come back if worse after 1 week"  
"No running for now"

## Things to ask about

"Ask about the thing with my knee"  
"What was that term he used?"  
"Check if this affects my other meds"



LIVE DEMO

**Let's see it running: [Click here](#)**

Fine-tuned handwriting model · field classifier · real sample document



# Organization, not diagnosis

- No autonomous diagnoses, treatment recommendations, or dosing decisions, ever
- Every structured output is labeled as a draft requiring the patient's or a clinician's review
- Built for organization and communication support, not as a substitute for medical judgment



**E-prescribing digitized the drug order.**

**CareCompanion digitizes the conversation around it.**

Thank you.

